TEACHING LIFE SCIENCES
IN THE AGE OF AI

Fast Tools, Slow Thinking

What building with AI taught me about where teaching is going



Chris Deppmann · UVA Biology

Illustrations are AI-generated.

PREDICT FIRST

How many US teens used an AI chatbot for schoolwork last fall?

OUT OF 100 TEENS, ABOUT HOW MANY?

About 15%

raise a hand

About 35%

raise a hand

More than half

raise a hand

Commit before you see anything.

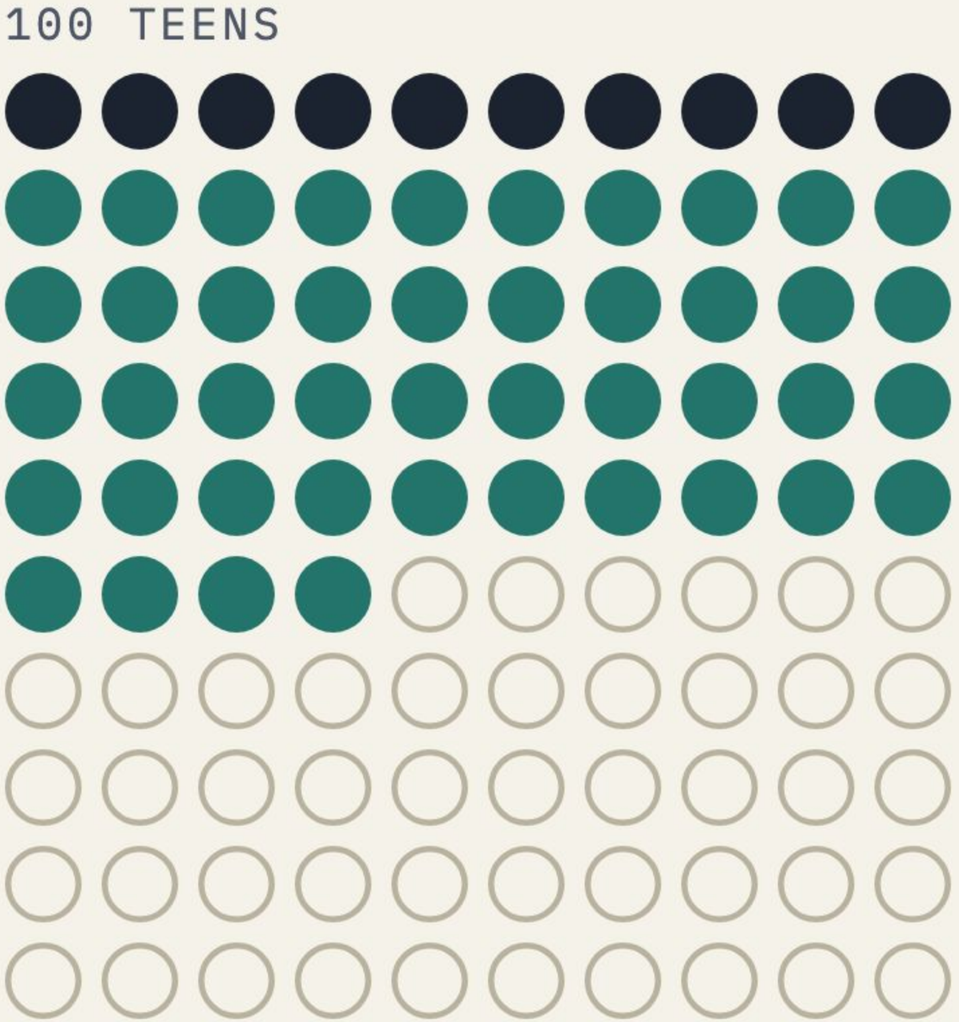
REVEAL

Your students already live here.

54%

of US teens used an AI chatbot for schoolwork, fall 2025.

- 1 in 10: all or most of their schoolwork
- The rest of the 54%: some schoolwork



Pew Research Center

WHERE I'VE LANDED

Fast tools for us. Slow thinking for them.

FAST TOOLS
· for teachers

Teachers are essential. Any AI tutor should be under the teacher's control, with every exchange visible to the teacher.

SLOW THINKING
· for students

Leaning on AI hurts students, especially young ones. They need slow thinking to make an idea their own.

THE ROOM YOU TEACH IN

And this is where you work.

49

hours a week

RAND, 2025

$895

of your own money, every year

AdoptAClassroom, 2025

68%

of teachers now run AI detectors on student work

CDT survey

22.6%

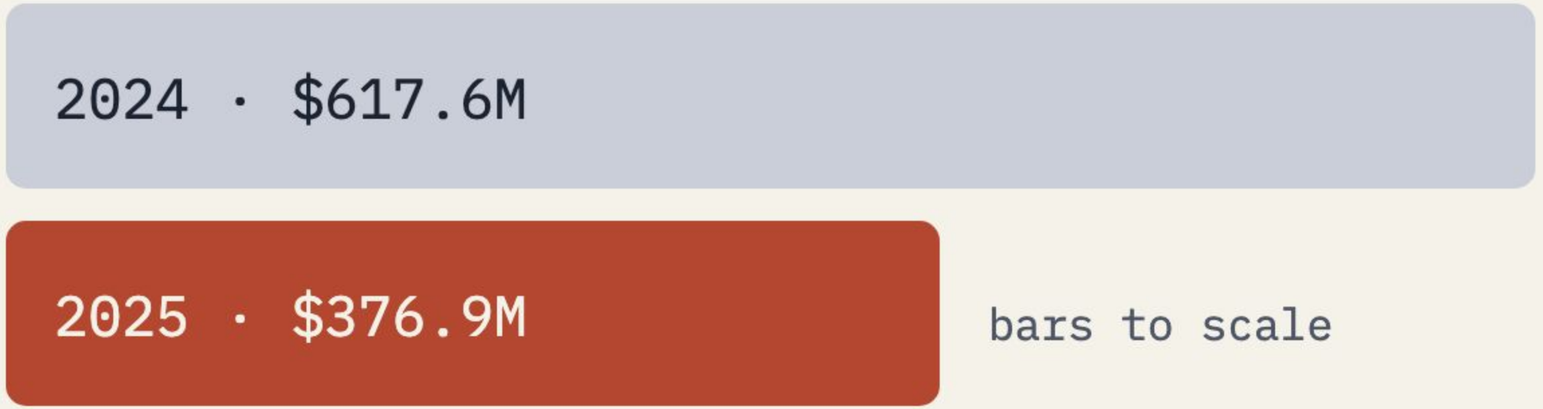
of US students chronically absent

2024–25 school year

WHAT JUST CHANGED

The answer just became free.

Chegg, a homework-answer company · yearly revenue



bars to scale

−39%

in one year, once students could use a chatbot

> College students pay about half what they did in 2007 for materials. The money moved from the book to the platform.

Chegg 10-K · NACS Student Watch · AAP StatShot

WHAT WE MEAN BY AI

Most of us still picture a chatbot. That picture is already out of date.

2022



A chatbot

You ask. It answers.

one turn at a time

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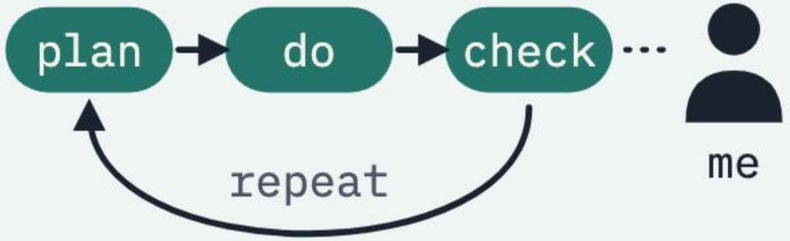


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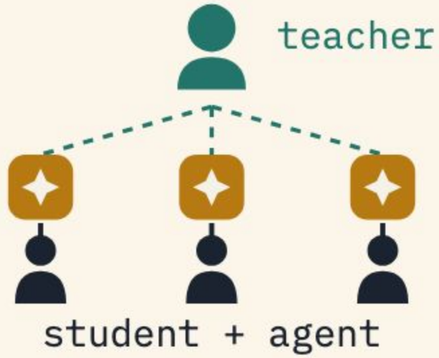
NOW



Agents

> How I built the site: agents drafted, agents fact-checked, I reviewed.

NEXT



An agent for every student

> **The teacher sees every exchange.**

a direction, not a date

PART 1 OF 2

I built it to find out

What happens when making materials stops being the hard part?



WHAT AGENTS CHANGED

The next 324 lessons took 13 days.

17 weeks



30 lessons · Jun 12 to Oct 10

13 days



bars drawn to scale

324 lessons · Oct 11 to 23

That’s the difference agents made.

Today: 354 lessons · 11 courses · 464 films · **free**

WATCH 60 SECONDS

Here’s the experiment.

biochemistrypedia.com

A switch

sets how much help.

The task stays the same.

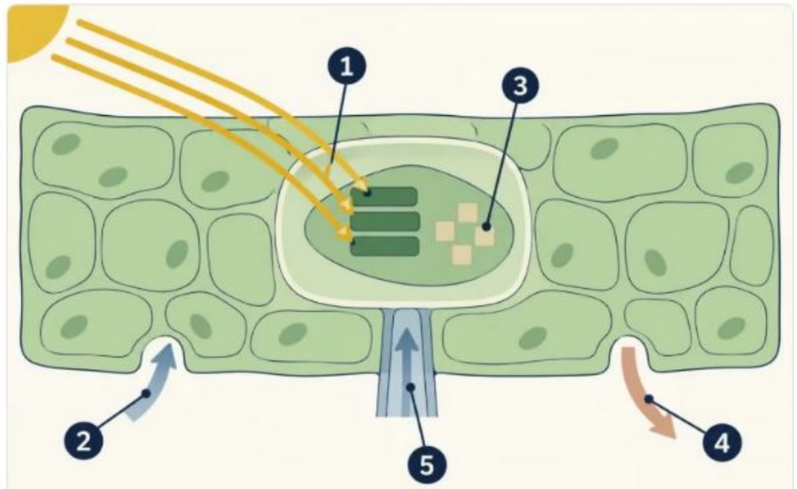
SUPPORTS

- ○ **Just the task** — No extra help
- ○ **Some help** — Word banks, partly labelled diagrams
- ◉ **Most help** — Adds sentence starters, fully labelled diagrams

Supports change how much help you get, not what you learn.

BLANK | PARTLY LABELLED | FULLY LABELLED

CAPTURING SUNLIGHT · BIO.2E



1 light 2 carbon dioxide 3 glucose 4 oxygen 5 water

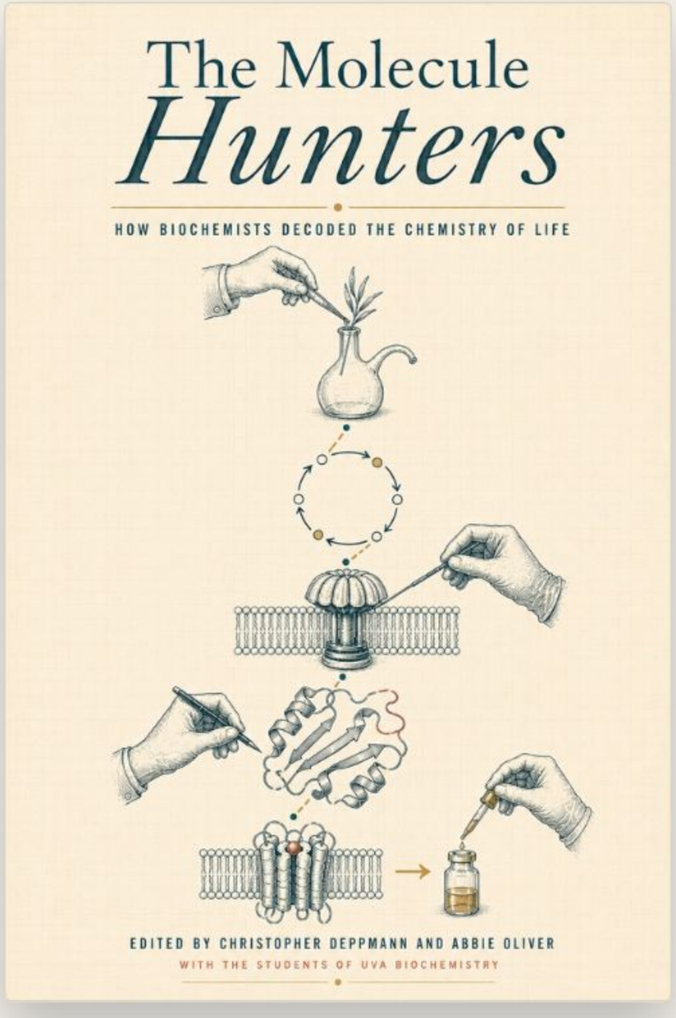
Inside one leaf cell, the chloroplast captures light and uses materials from the air and roots to make sugar and release a gas.

SAME TASK AT EVERY LEVEL

▶ 0:00

ANOTHER WAY I'VE USED AI

500 students wrote a book with me.



133 profiles

190 scientists

500+ UVA student co-authors

1 Every profile began as an AI deep-research dossier.

2 Free as a PDF, an EPUB and an audiobook.

3 Errors are listed in public, on a live fact-check page.

`The Molecule Hunters · deppmannlab.com/book`

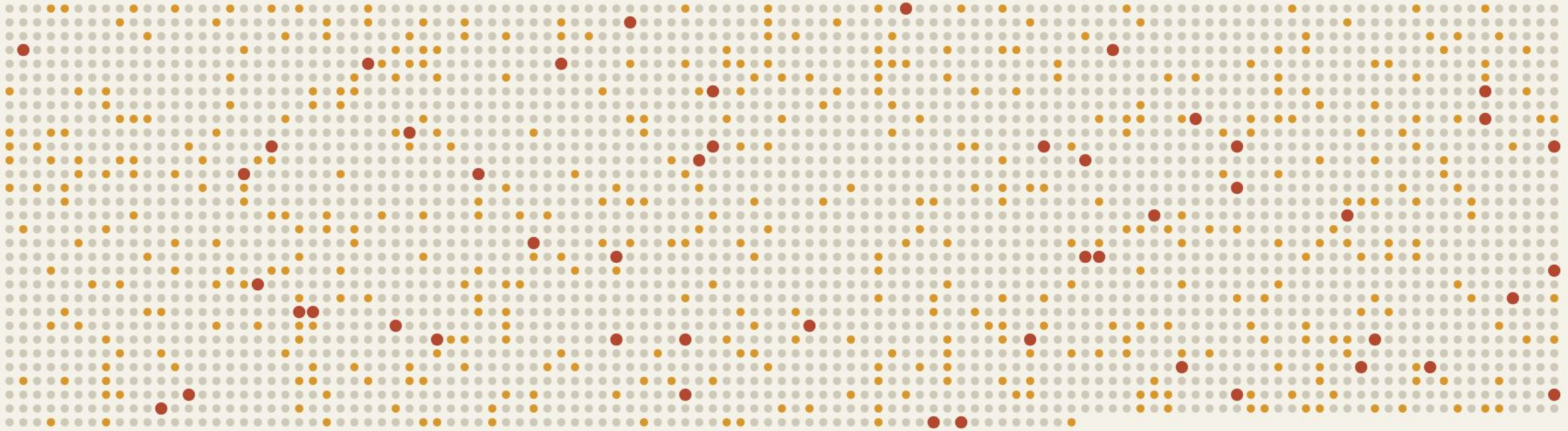
WHAT BUILDING TAUGHT ME

Each thing I built taught me something I didn't expect.

	I BUILT	IT TAUGHT ME
1	354 free lessons	→ Making is easy. Checking is the real work.
2	Pre-work students do before class	→ I could finally see how they think.
3	An AI to read their answers	→ It rewarded long answers, not good reasoning.
4	A place for their questions	→ Their best questions went unread.

Each answer raised the next question.

Making got cheap. Checking is the job.



49 flat wrong

453 imprecise

of 3,468 claims
in one course

> As we improved the process, wrong claims fell from **4 per lesson to under 1**.

Most of the work happens the night before. Now I can see it.

THURSDAY MORNING



2 finished

FRIDAY, 8:40 AM



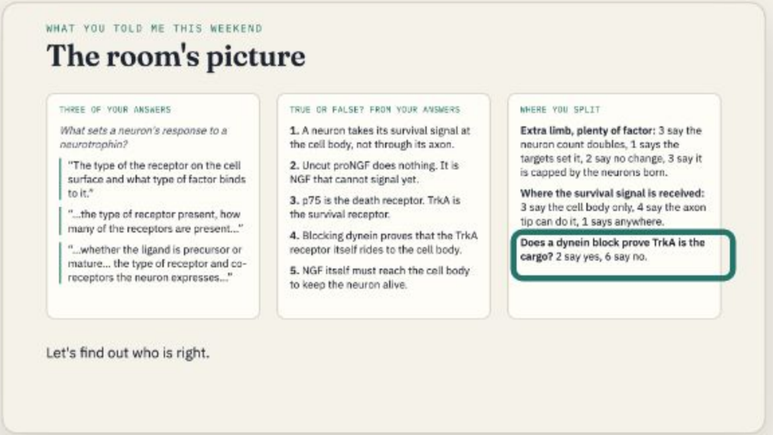
9 finished

+ 3 partly done

■ finished ◧ partly □ not yet

One grad course, 12 students
one 3–4 hour module

A REAL ROOM SLIDE



WHAT YOU TOLD ME THIS WEEKEND

The room's picture

THREE OF YOUR ANSWERS

What sets a neuron's response to a neurotrophin?

> "The type of the receptor on the cell surface and what type of factor binds to it."

> "...the type of receptor present, how many of the receptors are present..."

> "...whether the ligand is precursor or mature... the type of receptor and co-receptors the neuron expresses..."

TRUE OR FALSE? FROM YOUR ANSWERS

1. A neuron takes its survival signal at the cell body, not through its axon.
2. Uncut proNGF does nothing. It is NGF that cannot signal yet.
3. p75 is the death receptor. TrkA is the survival receptor.
4. Blocking dynein proves that the TrkA receptor itself rides to the cell body.
5. NGF itself must reach the cell body to keep the neuron alive.

WHERE YOU SPLIT

Extra limb, plenty of factor: 3 say the neuron count doubles, 1 says the targets set it, 2 say no change, 3 say it is capped by the neurons born.

Where the survival signal is received: 3 say the cell body only, 4 say the axon tip can do it, 1 says anywhere.

Does a dynein block prove TrkA is the cargo? 2 say yes, 6 say no.

Let's find out who is right.

Class opens with their own words: 2 say yes, 6 say no.

PREDICT FIRST

I asked an AI to score how deeply each student reasoned. Where did it rank the student who put in the most time?

ONE GRAD COURSE, 12 STUDENTS

Near the top

raise a hand

In the middle

raise a hand

Near the bottom

raise a hand

Commit before you see anything.

The student who put in the most time scored second from the bottom.

$r = 0.74$

Its scores tracked how much they wrote.

AI SCORE VS. ANSWER LENGTH

> **AI can read every answer. That doesn't mean it reads them fairly.**

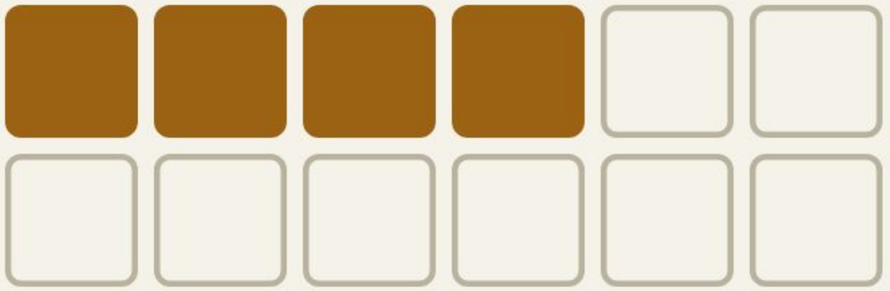
One grad course, 12 students · my own AI rater

About a quarter of their questions went past the lesson. Nobody read them.

WHAT THEY ASKED

- *a drug for this*
- *cancer*
- *dopamine and reward*

WHO ASKED



Four of 12 students asked most of them.

The system saved every question.

One grad course, 12 students

PART 2 OF 2

Where this is going

Five shifts that will look obvious in ten years.



The answer is free.
The trail is the evidence.

NOW

Blue-book sales are up **80%** at Berkeley.
Detectors flag non-native writers.

NEXT

How a student got there will count more than the answer.

Your students' AI reads the lesson before they do.

NOW

Unguarded GPT-4: practice up **48%**, exams down **17%**. A guarded tutor didn't.

NEXT

Lessons written for two readers: the student and their AI tutor — a tutor the teacher controls and can see.

Materials are easy now. Your attention is the scarce thing.

NOW

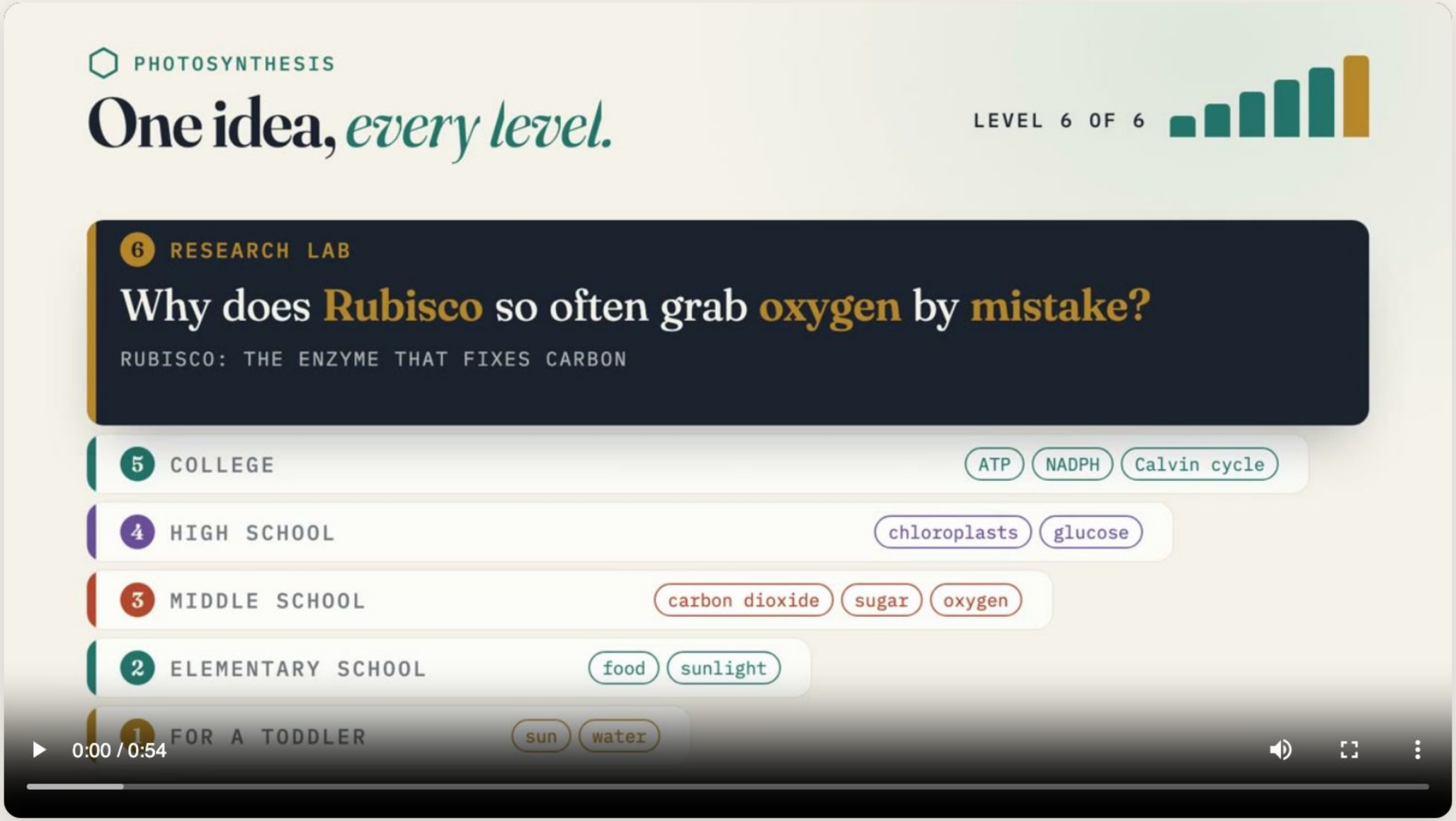
Teachers who use AI every week save about **5.9 hours** a week, mostly on prep.

NEXT

Each morning: which three students to see, and what to ask.

WATCH 55 SECONDS

One idea. Every level.



Meet every student where they are.

NOW

In one grade, students' levels vary widely. Software that taught each student at their own level raised math scores **0.36 SD** in 4.5 months (Delhi, 2019).

NEXT

Every student is checked as they go: sent to a review lesson when they need it, free to race ahead when they're ready, and the teacher is told the moment someone falls behind.

Their questions become the curriculum.

NOW

Interest in science falls from **9 to 17**. We teach to the test.

NEXT

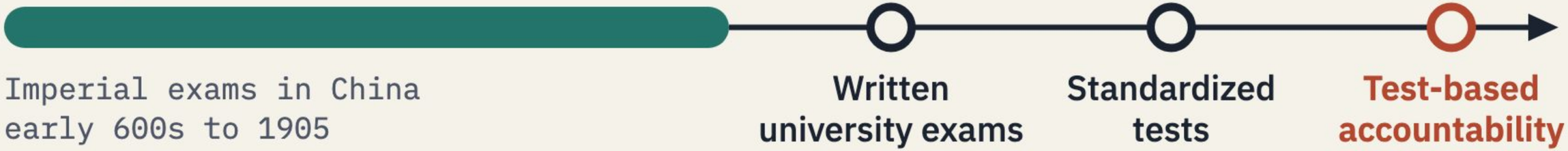
Their questions are part of the course. Exploring earns credit.

WHY WE TEACH TO THE TEST

When a measure becomes a target, it ceases to be a good measure.

Marilyn Strathern (1997), about exams, after Charles Goodhart (1975)

More than a thousand years of high-stakes exams



CHEAP TO MEASURE

The answer

Quick to grade.

teaching to the test

WHAT WE CARE ABOUT

The understanding

Slow to read.

HOW WE BREAK THE HABIT

When the answer is free, understanding is the only thing left worth grading.

> Goodhart's law bites hardest on **one narrow, high-stakes number**.

1



Measure the thing itself.

The reasoning, not a stand-in for it.

2



Many small looks, not one big test.

Low stakes, so nothing is worth gaming.

3



A person reads it.

Judgment, not just a score.

HONESTLY State tests and the MCAT aren't going away soon. If understanding is what we look at every day, the test becomes a by-product.

HONEST ABOUT THE RISKS

It can go wrong.

1. Free tools help the already-engaged most.
2. In one 2026 study, a course AI tutor lowered grades, most for first-generation students.
   Not yet peer-reviewed
3. A record of how a child thinks is sensitive.

THE RULE

AI reads. You decide.

WHERE I LAND

Fast tools for us. **Slow thinking** for them.

What I'd ask of any AI in a classroom

1. **The student tries first.**
   It never gives the answer before they commit.
2. **The teacher sees every exchange.**
3. **It meets each student at their level.**
   And tells the teacher who is falling behind.
4. **It looks at reasoning.**
   And gives second chances.
5. **Everything it makes is checked.**
   Mistakes are fixed in public.

We can do better.

My hope is that this is an enabling technology.

biochemistrypedia.com
/teachers